



THE UNIVERSITY OF
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FINAL REPORT

**The Art of Scientific Computing:
Project Title: The Bak–Tang–Wiesenfeld
Sandpile Model on a 3×3 Lattice**

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Chapter 1

Bak–Tang–Wiesenfeld Sandpile Model

1.1 Project aim

This project implements and studies the Bak–Tang–Wiesenfeld (BTW) sandpile model on a finite 3×3 square lattice. The main aim is to create a correct simulation and use it to examine the long-term behaviour of a small dissipative sandpile.

The project considers the following questions:

1. How does the mean height change from an initially empty lattice?
2. When does the system reach a statistically stationary state?
3. What are the distributions of avalanche topples, area, boundary loss and length?
4. How strongly are these avalanche measurements related?
5. How does the open boundary affect corner, edge and centre sites?

The 3×3 lattice is small enough for individual avalanches to be checked by hand. However, it also has strong finite-size effects. Therefore, the results will not be presented as evidence of a universal power law.

1.2 Model and method

The model uses a 3×3 lattice with open boundaries. Each site has an integer height z_{ij} representing its number of grains. The simulation begins with an empty lattice.

At each time step, one grain is added to a uniformly selected random site. A site becomes unstable when its height reaches four. An unstable site topples according to

$$z_{ij} \rightarrow z_{ij} - 4. \tag{1.1}$$

One grain is transferred to each nearest neighbour. Grains directed outside the lattice leave the system. Toppling continues until all site heights are below four.

For each avalanche, the program records:

- **topples**: the total number of toppling events;
- **area**: the number of distinct sites that topple;
- **loss**: the number of grains lost through the boundary;
- **length**: the maximum Euclidean distance from the grain-addition site to a site that topples.

The main simulation uses 10,000 grain additions. The first 100 steps are treated as a burn-in period and are excluded from the stationary statistics. Random seeds 42 and 43 were used to check whether the main conclusions depend strongly on a particular random sequence.

1.3 Implementation and validation

The original skeleton code has been developed into a working `SandPile` class. The program can create the lattice, add grains, topple unstable sites, calculate boundary loss and process a connected avalanche until the grid is stable.

The implementation was tested using hand-calculated examples. These included:

- toppling at the centre, an edge and a corner;
- a connected avalanche involving several sites;
- repeated toppling of the same site;
- two valid toppling orders applied to the same configuration;
- a check that every final grid is stable; and
- a check of mass balance.

For example, after 1,000 grain additions, one test simulation had a final mass of 16 and a total boundary loss of 984. Therefore,

$$M_{\text{input}} = M_{\text{final}} + M_{\text{loss}} = 16 + 984 = 1000. \quad (1.2)$$

The two different valid toppling orders also produced the same final stable configuration.

1.4 Current results

After the initial transient, the mean height fluctuated around a stable level rather than continuing to increase. The stationary mean heights were 1.8150 for seed 42 and 1.8155 for seed 43. This close agreement suggests that the result is not strongly dependent on one random sequence.

Quantity	Seed 42	Seed 43
Stationary mean height	1.8150	1.8155
Height standard deviation	0.2482	0.2536
Recorded stationary avalanches	3214	3201
Avalanche probability	0.3246	0.3233
Mean boundary loss per step	0.9996	1.0004

Table 1.1: Comparison of the two 10,000-step simulations.

The mean boundary loss is approximately one grain per stationary step. Since one grain is also added per step, the average input and output are approximately balanced. This provides evidence that the system has reached a statistically stationary state.

For seed 42, the mean avalanche measurements were 2.525 topples, 2.518 distinct toppled sites, 3.079 lost grains and a length of 0.866. The largest recorded avalanche had 10 topplings, covered all nine sites, lost 12 grains and had a maximum length of $2\sqrt{2}$.

The avalanche measurements were strongly positively correlated. The correlation between topples and area was 0.999, while the correlations between topples and loss and between topples and length were 0.933 and 0.877 respectively. The very strong topples–area relationship is partly caused by the small lattice, where most sites topple at most once during an avalanche.

The open boundary also produced a spatial pattern. For seed 42, the mean heights were 1.748 at corner sites, 1.851 at edge sites and 1.937 at the centre. Corner sites have the lowest mean height because a corner toppling loses two grains directly. An edge toppling loses one grain, while a centre toppling has no direct boundary loss.

The program has produced plots of the mean height, avalanche distributions, correlations and the spatial mean-height pattern. These figures will be selected and added to the final version of the report.

1.5 Reproducibility and version control

The simulation has been reorganised into a parameterised function so that the number of steps, burn-in period and random seed can be changed. Numerical results are saved in NumPy `.npz` format. A separate plotting script can load the saved data and reproduce the figures without rerunning the complete simulation.

The report template was forked from the course GitHub repository. The report is currently being edited in the forked project. The updated LaTeX files, figures, bibliography and compiled PDF will be committed to the fork before a pull request is submitted to the original course repository.

1.6 Remaining work

The next tasks are:

The immediate next step is to select the most useful figures and write clear captions for them. The numerical results have already been saved in a machine-readable NumPy file, so the reported values can be checked and reproduced.

The report will then be completed by adding a discussion of statistical stationarity, dissipation at open boundaries, relationships among avalanche measurements and the limitations of the small lattice. Exact state-space enumeration or a modified model will only be attempted if the core report is complete.